

MATH3881/5881: Statistical Machine Learning Theory

Week 4: Theory Tutorial: Backpropagating the Variance

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In Week 3 we learned how backpropagation sends a gradient signal $\delta^{[\ell]}$ backwards through the layers of a network. For training to succeed, the magnitude of $\delta^{[\ell]}$ at every layer must be reasonable, neither vanishingly small (the early layers will never move) nor explosively large (the updates become unstable).

This tutorial asks: in a deep ReLU network with random initial weights, how does $\|\delta^{[\ell]}\|$ change as ℓ decreases from L down to 1? We will derive a clean recursion for its expected squared magnitude, and then read off the unique scaling of the initial weights that keeps the recursion stable for any depth L . The answer is the He criterion, i.e., the variance of each weight $s_W^2 = 2/n$, which derived here from a backward-pass argument. The factor of 2 in He's formula will pop out automatically, with a clear explanation: ReLU silences half of the gradient at each layer, and He compensates by doubling the weight variance.

Background

For your convenience, below is the background concepts from lecture notes that are required for solving the problems.

The network. Consider an L -layer feedforward network with the same width n at every layer, no bias, and ReLU activations. With input $x \in \mathbb{R}^n$, weight matrices $W^{[\ell]} \in \mathbb{R}^{n \times n}$, pre-activations $z^{[\ell]}$ and activations $a^{[\ell]}$, the forward pass is

$$a^{[0]} = x, \quad z^{[\ell]} = W^{[\ell]} a^{[\ell-1]}, \quad a^{[\ell]} = \sigma_{\text{ReLU}}(z^{[\ell]}) := \max(z^{[\ell]}, 0) \quad (\text{element-wise}),$$

for $\ell = 1, \dots, L$, and the loss is the squared error $C = \frac{1}{2} \|a^{[L]} - y\|^2$.

Backprop (Week 3). Writing $\delta^{[\ell]} := \partial C / \partial z^{[\ell]} \in \mathbb{R}^n$ and $D^{[\ell]} := \text{diag}(\dot{\sigma}_{\text{ReLU}}(z^{[\ell]})) \in \mathbb{R}^{n \times n}$, the standard chain-rule recursion is

$$\delta^{[L]} = a^{[L]} - y, \quad \delta^{[\ell]} = D^{[\ell]} (W^{[\ell+1]})^\top \delta^{[\ell+1]} \quad \text{for } \ell = L-1, \dots, 1.$$

Each diagonal entry $D_{jj}^{[\ell]} = \dot{\sigma}_{\text{ReLU}}(z_j^{[\ell]}) = \mathbf{1}[z_j^{[\ell]} > 0] \in \{0, 1\}$ is the gating indicator of neuron j : it is 1 if that neuron was active on the forward pass, and 0 otherwise. So at each backward step, the gradient is first projected onto the active neurons by $D^{[\ell]}$ and then linearly mapped by $(W^{[\ell+1]})^\top$.

Random initialisation. At $t = 0$, draw each weight entry $w_{ij}^{[\ell]}$ independently from a distribution with mean zero and variance s_W^2 . Independence is across all rows, columns, and layers. Treat the output error $\delta^{[L]} = a^{[L]} - y$ as a fixed vector for the analysis; we trace how the backward chain rescales it.

Working assumption. When taking expectations through the backward chain at layer ℓ , treat the three pieces, namely the weights $W^{[\ell+1]}$, the gate $D^{[\ell]}$, and the incoming gradient $\delta^{[\ell+1]}$, as mutually independent. This is the standard simplification that makes the variance recursion derivable in closed form.

Two facts we will use.

- *Vanishing cross-terms.*

$$\mathbb{E}[w_{ij}^{[\ell+1]} w_{kj}^{[\ell+1]}] = \begin{cases} s_W^2, & i = k, \\ 0, & i \neq k. \end{cases}$$

- *Active-neuron rate at initialisation.* Each pre-activation $z_j^{[\ell]} = \sum_i w_{ji}^{[\ell]} a_i^{[\ell-1]}$ has a distribution symmetric about 0 (flipping the sign of $W^{[\ell]}$ produces the same distribution of $z_j^{[\ell]}$), so $\mathbb{P}(z_j^{[\ell]} > 0) = 1/2$, and

$$\mathbb{E}[D_{jj}^{[\ell]}] = \frac{1}{2} \quad (\text{each ReLU neuron fires with probability } 1/2).$$

Quantity of interest. Define

$$V_\ell := \mathbb{E}[\|\delta^{[\ell]}\|^2], \quad \ell = 1, 2, \dots, L,$$

the expected squared magnitude of the backprop gradient at layer ℓ , where the expectation is over both the random weights and the activation gates the gradient passes through. Note $V_L = \|\delta^{[L]}\|^2$ (deterministic). We ask: how does V_ℓ evolve as we move backwards from layer L to layer 1?

Two problems

Problem 1: The variance recursion (“variance backprop”)

Show that for every $\ell = L - 1, L - 2, \dots, 1$,

$$V_\ell = \frac{n s_W^2}{2} V_{\ell+1},$$

and conclude by iteration that

$$V_1 = \left(\frac{n s_W^2}{2} \right)^{L-1} \|\delta^{[L]}\|^2.$$

Hint. The backprop equation gives

$$\delta_j^{[\ell]} = D_{jj}^{[\ell]} \sum_{i=1}^n w_{ij}^{[\ell+1]} \delta_i^{[\ell+1]}, \quad \text{so} \quad \|\delta^{[\ell]}\|^2 = \sum_{j=1}^n D_{jj}^{[\ell]} \left(\sum_{i=1}^n w_{ij}^{[\ell+1]} \delta_i^{[\ell+1]} \right)^2,$$

where we used $(D_{jj}^{[\ell]})^2 = D_{jj}^{[\ell]}$ (because $D_{jj}^{[\ell]} \in \{0, 1\}$). Condition on $\delta^{[\ell+1]}$ and apply the Background's working assumption (independence of $W^{[\ell+1]}, D^{[\ell]}, \delta^{[\ell+1]}$). Take the conditional expectation in two steps:

- (i) Expectation over $W^{[\ell+1]}$ (with $\delta^{[\ell+1]}$ and $D^{[\ell]}$ held fixed): apply the *vanishing cross-terms* fact from the Background. The result is $s_W^2 \|\delta^{[\ell+1]}\|^2 \sum_j D_{jj}^{[\ell]}$.
- (ii) Expectation over $D^{[\ell]}$ (with $\delta^{[\ell+1]}$ held fixed): apply the *active-neuron rate at initialisation* fact from the Background to get $\mathbb{E}[\sum_j D_{jj}^{[\ell]}] = n/2$.

Combine to read off the recursion.

Problem 2: The unique stable scaling

For the gradient signal to remain usable on a network of any depth L , the gradient at the first layer must remain comparable in magnitude to the gradient at the last layer, i.e. neither $V_1/V_L \rightarrow 0$ nor $V_1/V_L \rightarrow \infty$ as L grows. Show that this forces

$$s_W^2 = \frac{2}{n}.$$

This is the **He criterion** for ReLU networks. The factor of 2 comes directly from the fact that, on average, only half of the neurons at each layer pass the gradient through; He compensates by doubling the weight variance relative to the linear case.

Hint. From Problem 1, $V_1/V_L = (ns_W^2/2)^{L-1}$. Analyse the three cases $ns_W^2/2 > 1$, $ns_W^2/2 < 1$, and $ns_W^2/2 = 1$ as $L \rightarrow \infty$.

Take-home message

The variance of the backprop gradient propagates through the same chain of weights as the gradient itself: with iid weights and ReLU gating, each backward layer multiplies it by $ns_W^2/2$. The unique initialisation that keeps this factor at 1, so gradients neither vanish nor explode as the network deepens, is

$$s_W^2 = \frac{2}{n} : \quad \text{the **He criterion**}. \quad$$

The factor of 2 is structural: ReLU silences half the gradient at each layer, so He doubles the weight variance to compensate. The same criterion arises in Week 2 from forward-pass signal preservation, so He initialisation has two independent motivations, both delivering the same answer.

Initialisation vs. training

The recursion above is a statement about the initialisation (i.e., $t = 0$) only. Once training begins, gradient updates correlate the weight entries with the data, the pre-activations $z_j^{[\ell]}$ stop being symmetric about 0, and the active-neuron rate $\mathbb{E}[D_{jj}^{[\ell]}] = 1/2$ no longer holds: both Background facts break. What He initialisation guarantees is that the network *starts* in a trainable state. Modern architectures and optimisers then keep it trainable as the iterations proceed, each one addressing one of the assumptions the proof relied on:

- **Batch / Layer Normalisation** re-centres and re-scales the pre-activations at every iteration. This stops the active-neuron rate $\mathbb{E}[D_{jj}^{[\ell]}]$ from drifting toward 0 (the “dead ReLU” regime, where the Background fact fails catastrophically and the gradient at that layer vanishes); it does not pin the rate at exactly 1/2, since BN’s learnable scale and shift (γ_j, β_j) are free to move during training, but the catastrophic drift is suppressed.
- **Adaptive optimisers** (Adam, RMSProp from Week 4) rescale the update coordinate-wise, so a small V_ℓ at the early layers is compensated by a larger effective step size.